

GIS-BASED FLOOD RISK ASSESSMENT OF FIRE ISLAND, NY: A LIDAR-DERIVED DEM ANALYSIS FROM ROBERT MOSES TO WATCH HILL.



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Abstract:

As climate change accelerates, rising sea levels and increasingly intense coastal storms have contributed to more frequent and severe flooding in low-lying coastal regions. Fire Island, a barrier island off the southern coast of Long Island, New York, is especially susceptible to flooding due to its narrow width, minimal elevation, and exposure to both the Atlantic Ocean and Great South Bay. The study area, extending from Robert Moses Beach to Watch Hill, is characterized by low relief topography and dynamic coastal processes that increase its vulnerability to storm surge and tidal inundation.

To evaluate potential flooding scenarios within this region, Geographic Information Systems (GIS) techniques were employed to model and analyze projected inundation patterns. High-resolution Light Detection and Ranging (LiDAR) data were processed using Global Mapper to generate a bare-earth Digital Elevation Model (DEM), removing vegetation, structures, and other surface features to accurately represent ground elevation. This elevation model enabled detailed flood simulations under varying sea-level rise scenarios and storm surge conditions.

Results from the simulations indicate that substantial portions of the study area, particularly low-elevation bayside zones and interior regions closer to the Atlantic shoreline and low-lying interior zones, are at significant risk of inundation. Areas with slightly higher dune elevations provide limited natural protection, though these features are vulnerable to erosion and overwash during major storm events. The findings highlight priority zones for emergency planning and resource allocation, particularly the west end access point of the Robert Moses Causeway.

The GIS-based modeling framework utilized in this study can be adapted for other barrier island systems and coastal environments worldwide. By identifying areas of highest flood susceptibility, this approach supports coastal management strategies, infrastructure planning, and emergency response preparedness in the face of continued sea-level rise.

Background Information:

- Fire Island is a barrier island located off Long Island's south shore(Fire Island Association, n.d.)
- Fire Island is home to 17 residential communities(Fire Island Association, n.d.).
- Fire Island is primarily accessible by nautical transportation via ferry or personal watercraft with limited vehicular access.
- Vehicular access is restricted to two bridges Robert Moses Causway (west) and Smith's Point Bridge (east).
- Due to climate change, there has been an increase in the severity and frequency of storms, which has led to an increase in coastal flooding.

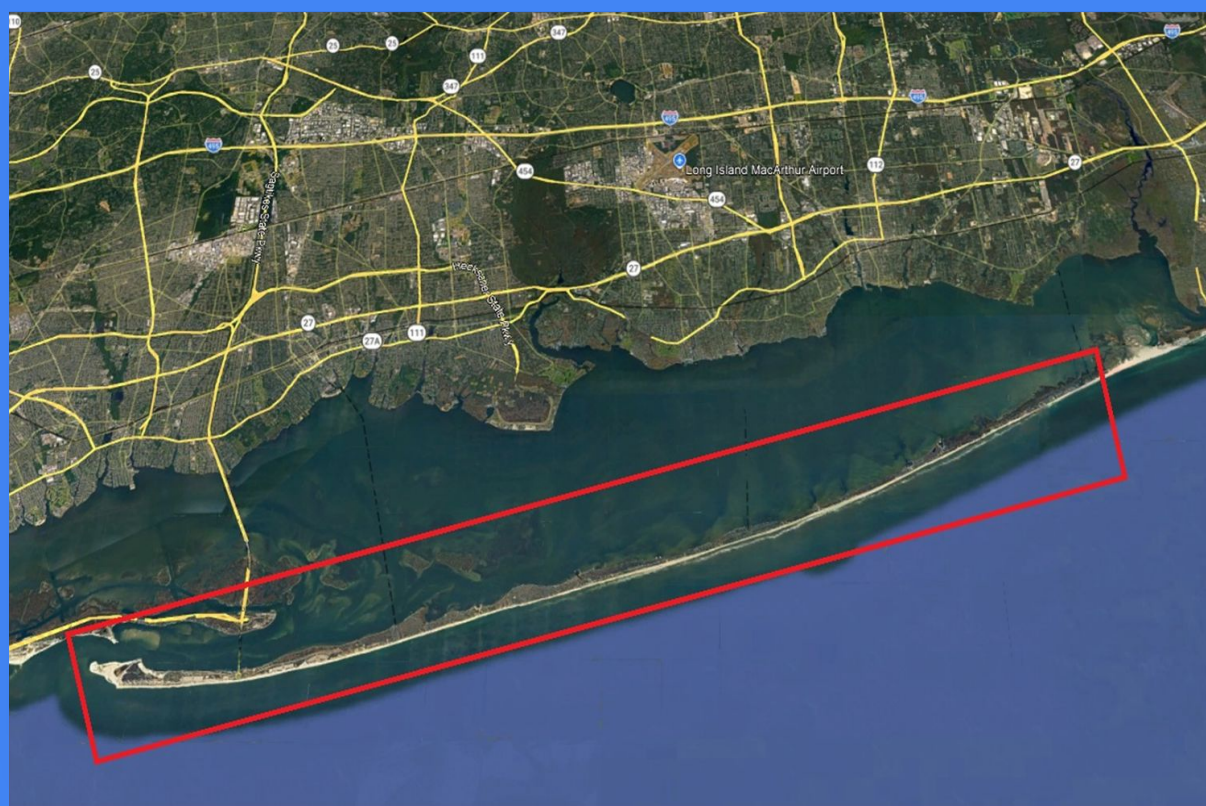


Figure 1: Map of Fire Island Study Area, From Robert Moses to Watch Hill.

Methods:

- Create polygon in Google Earth → Global Mapper
- Generate bare-earth DEM (10 m NED, island-wide)
- Create high-resolution DEM (LiDAR, Robert Moses Bridge)
- Apply bathtub inundation model (hydraulic connectivity)
- Model water levels: +1 m, +2 m, +3 m, +5 m (MSL)
- Map and analyze flood extent patterns

Results:

•Low-lying bayside regions and interior sections between Robert Moses Beach and Watch Hill are among the first areas to become inundated. (1–2 m).

•Higher elevation dune systems along the Atlantic shoreline remain relatively protected during initial stages of flooding but become increasingly vulnerable at higher water levels (3–5 m)

•At approximately 1–2 m of water elevation, sections of low-lying access roads begin to experience inundation

•As water levels increase to 2–3 m, connectivity along the island becomes increasingly fragmented

•The analysis indicates that under moderate to high water level scenarios (approximately 2–3 m), portions of the approach to the causeway and surrounding infrastructure are at risk of flooding.

• At higher water levels (3–5 m), access to and from Long Island via the bridge is likely to be significantly disrupted or entirely compromised.

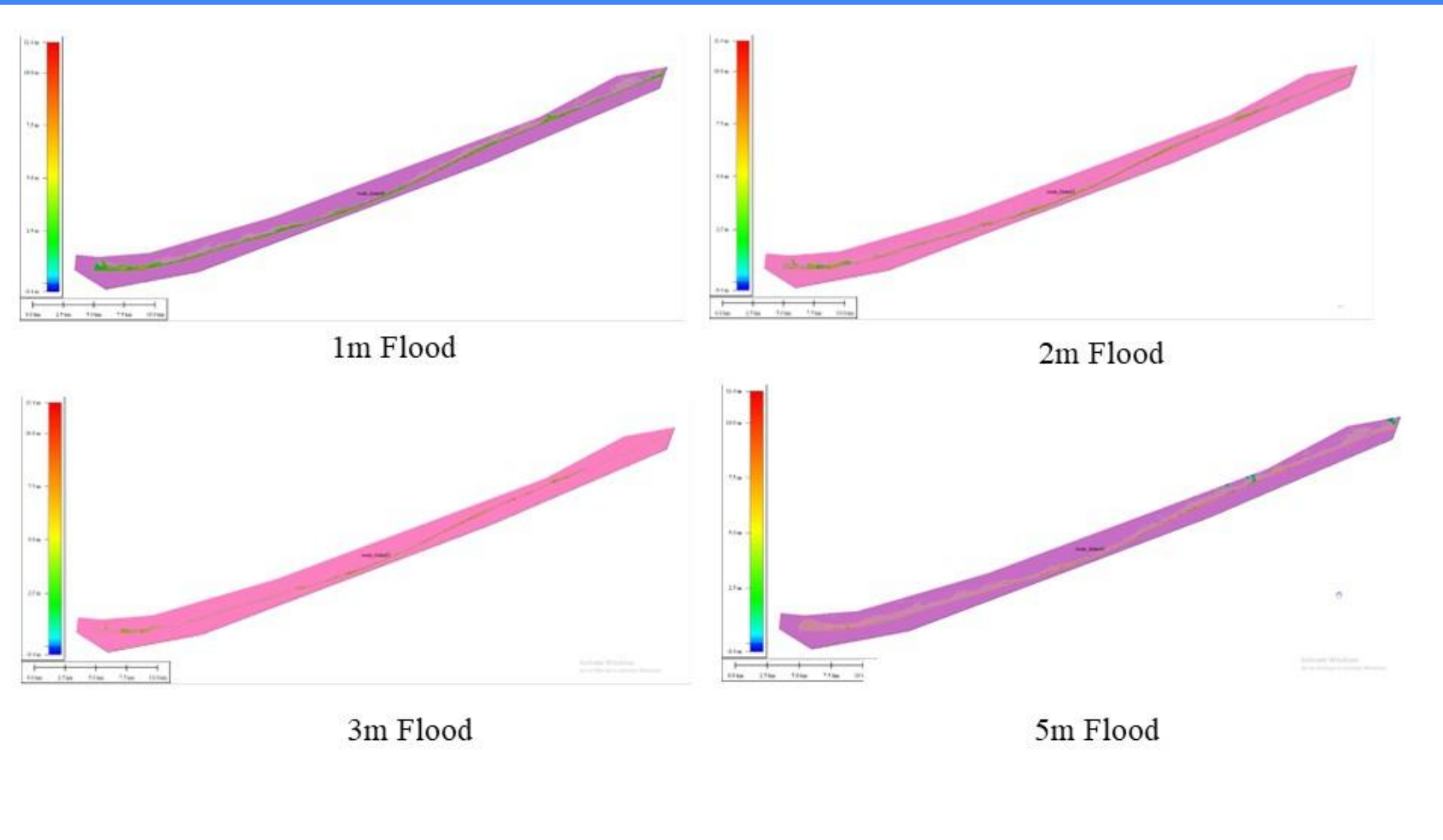


Figure 2: Island-Wide Analysis, NED 10m Res., 1-5m. Flood Simulation.

Table 1:Total flooded area (km²) and percentage of inundation across Fire Island under incremental water level scenarios (1–5 m) based on bathtub model simulations.

m	Total area flooded (sq.km)	Percentage (%)
0	0	
1	11.59	45.0
2	17.31	67.1
3	21.04	81.6
4	23.02	89.3
5	24.4	94.6
Total area	25.78	

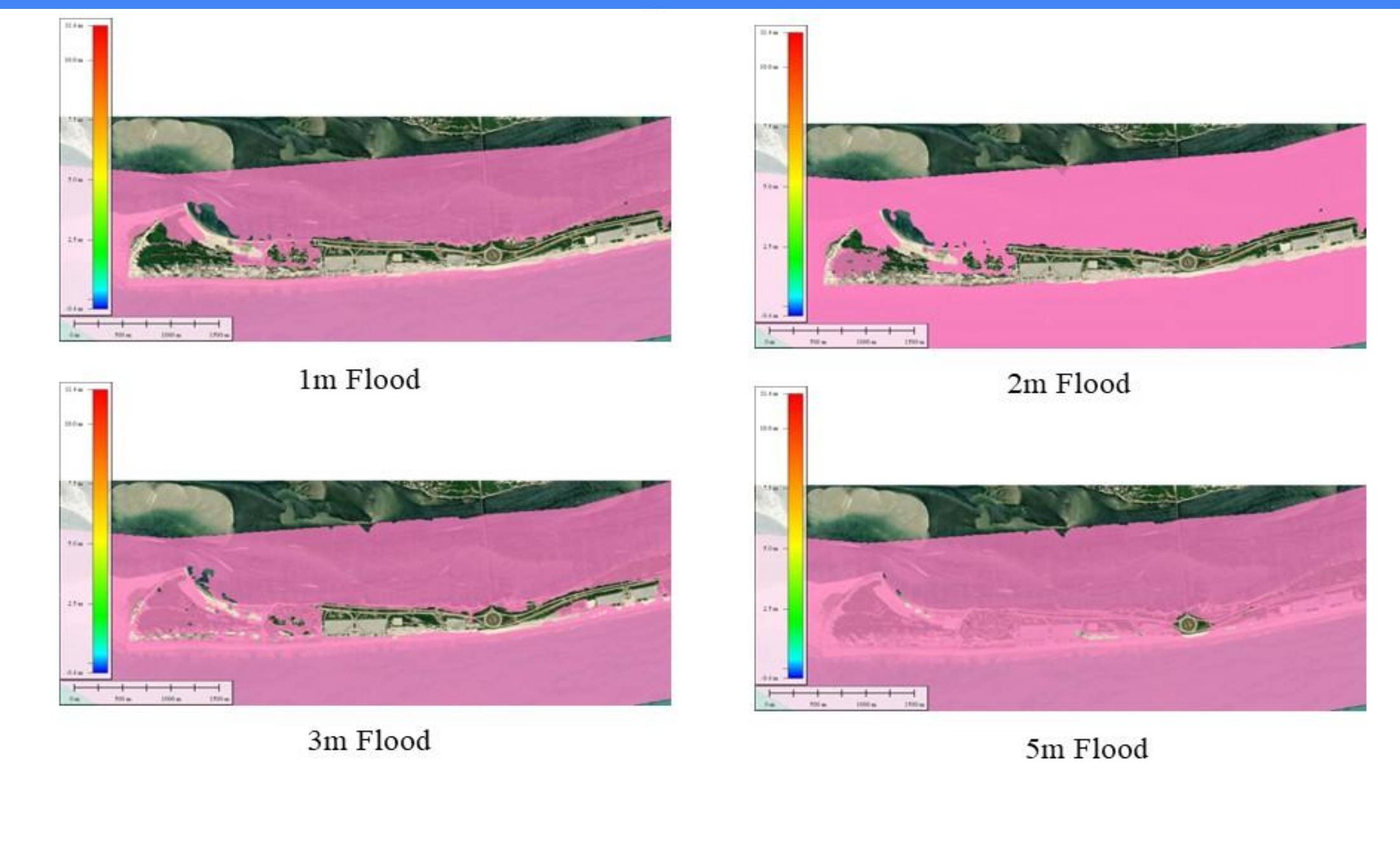


Figure 3: Focused Analysis of Robert Moses Bridge, NED 10M Res., 1-5m flood simulation.

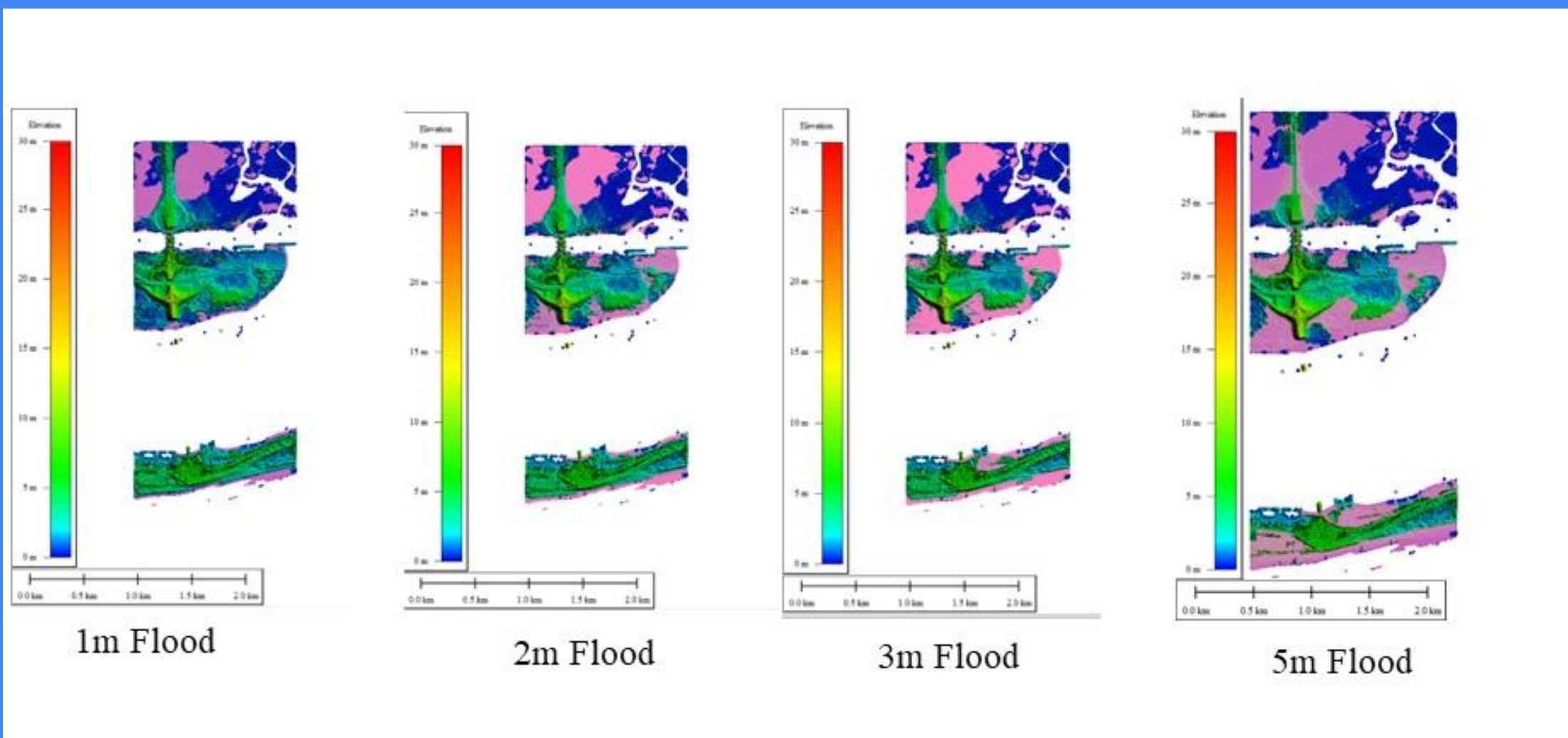


Figure 4: Focused Analysis of Robert Moses Bridge, LiDAR dataset, 1-5m flood simulation.

Discussion:

- The bathtub model shows that even modest water level increases cause early flooding in low-lying interior and bayside areas
- Road network fragmentation—especially along the central corridor and near the Robert Moses Causeway—leads to loss of accessibility at ~2–3 m
- Future work should extend LiDAR-based modeling to the Smith Point Bridge (eastern access point) using high-resolution DEMs and 1–5 m water level scenarios
- Integrating both access points will improve understanding of evacuation feasibility and identify thresholds where the island becomes fully inaccessible

Conclusion:

- Low-lying areas flood under modest water levels (1–2 m), with dunes offering limited protection in higher scenarios.
- Infrastructure fails before full inundation, as road networks fragment and access—especially near the Robert Moses Causeway—is lost (~2–3 m).
- The bathtub inundation model, combined with high-resolution DEMs, effectively simulates flooding in low-relief coastal environments.
- Findings support coastal management and infrastructure planning and provide a transferable framework for similar regions.

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Credit Authorship Contribution Statement:

Runkel, M.: Writing- Abstract, Intro, Methods, Results, Discussion, Conclusion, and References.editing, literature, global mapper flood simulations, Figure 1, Figure 2, Figure 3, Figure 4, Figure 5.
Marsellos, A.E.: supervision, guidance, editing.

References:

FI Communities — Fire Island Association. (2024, July 1). Fire Island Association.
<https://www.fireislandassociation.org/fi-communities>

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